TOPIC – LINUX

Assignment 1: Practice on Commands

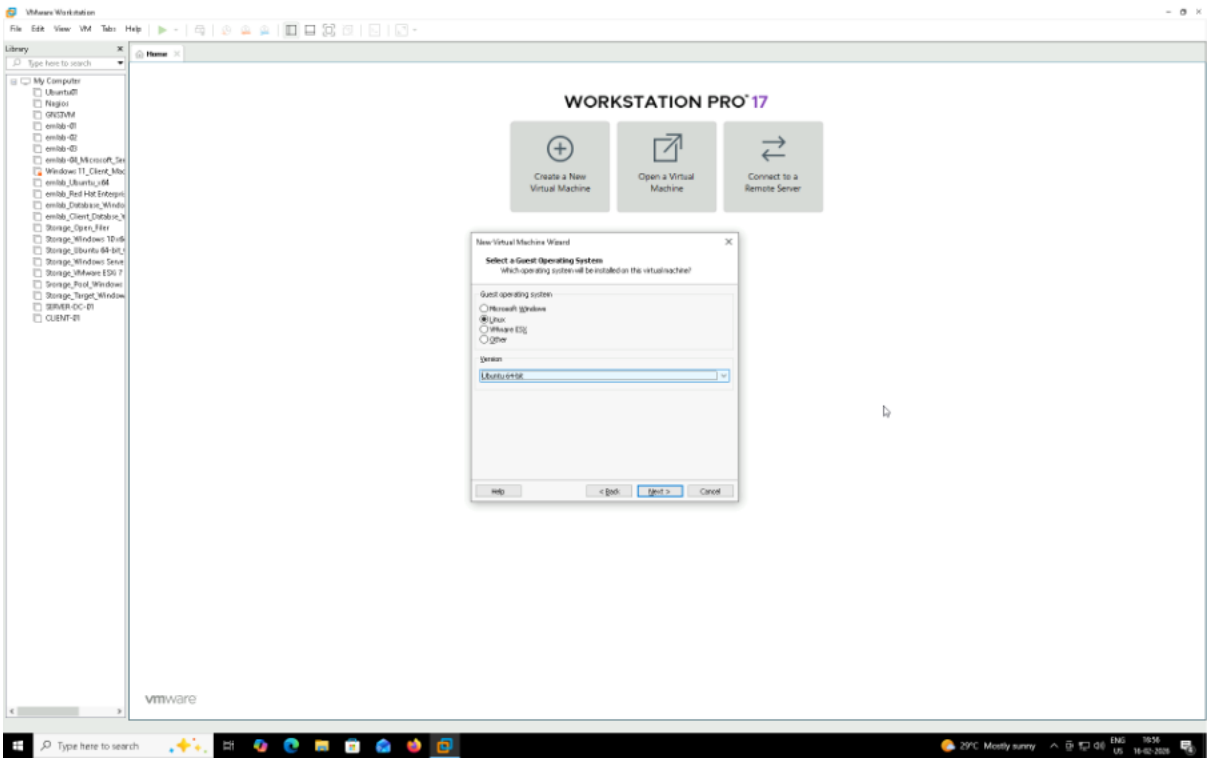
Chmod, chown ,find ,grep

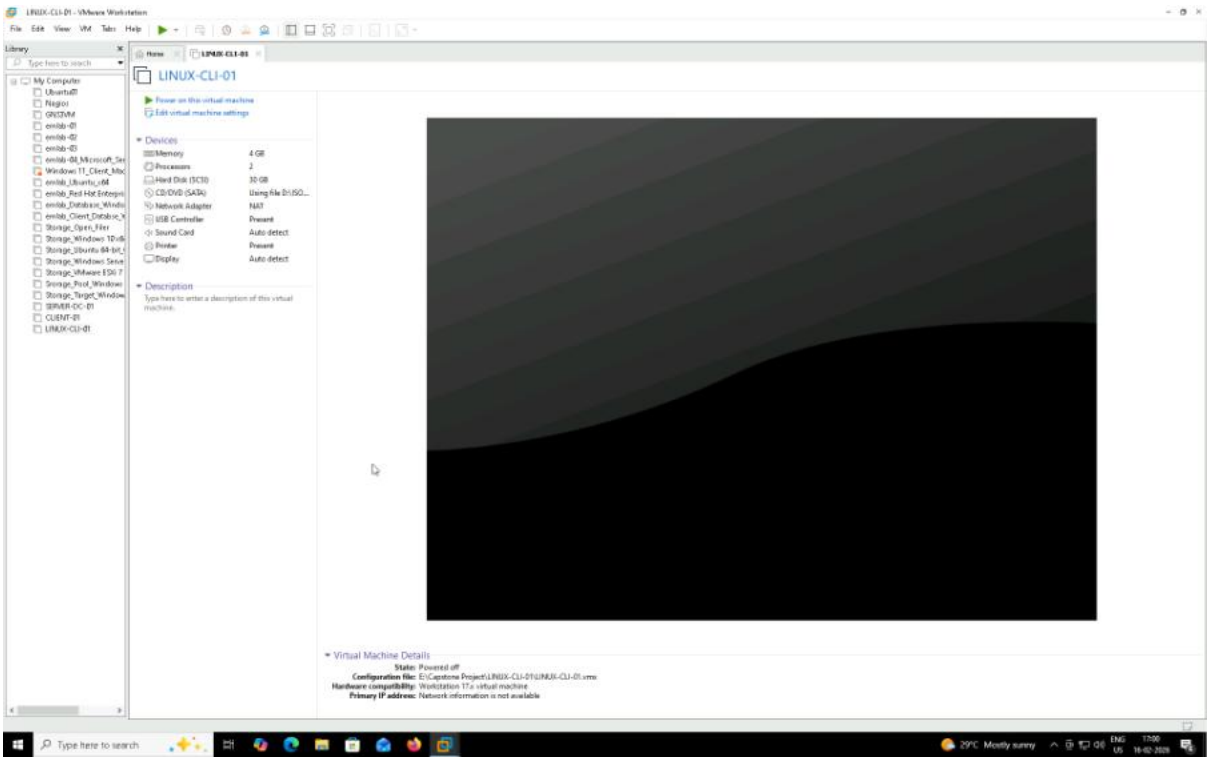
As part of this assignment, you need to complete below tasks:

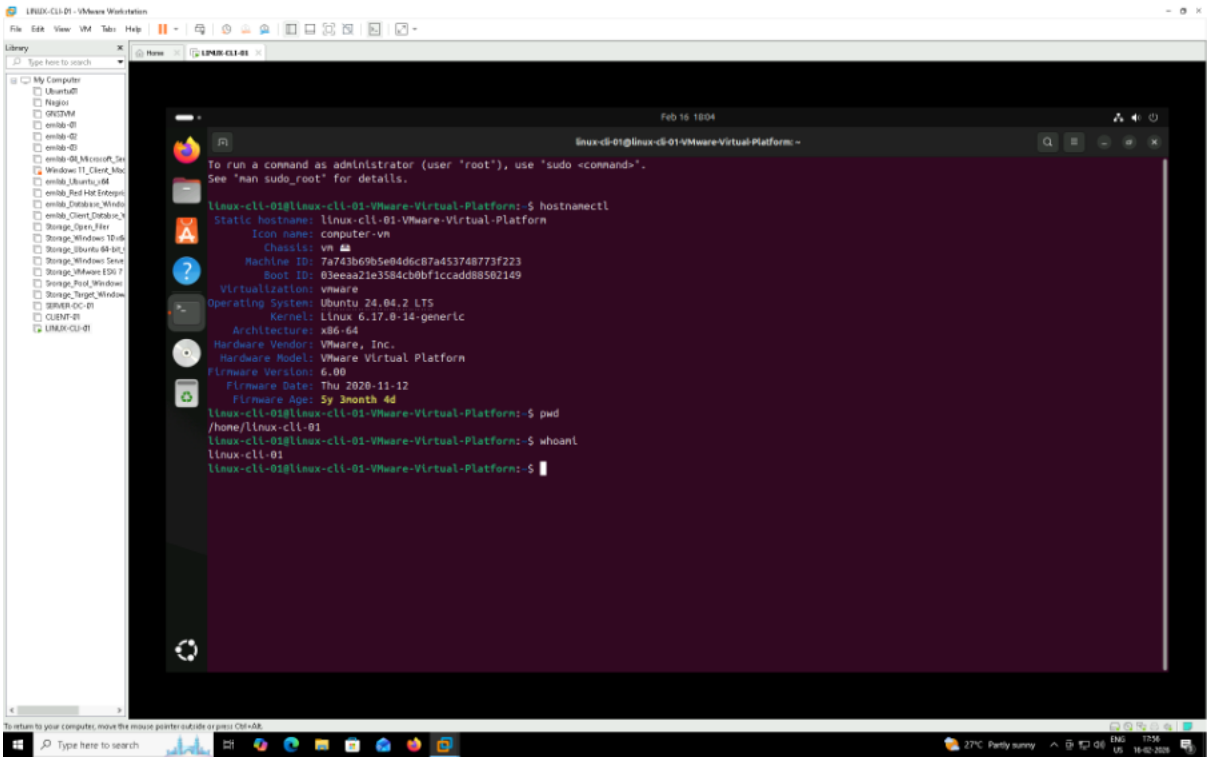
1. Create personal project directory
2. Make subdirectories
3. Create text files
4. Copy and move files
5. Understand file permissions

We will now create an Ubuntu x64 VM to accomplish the tasks: -

- Name: - LINUX-CLI-01
- OS: - Ubuntu Linux (64-bit)
- CPU: - 2 vCPU
- RAM: - 4 GB
- Disk: - 30 GB

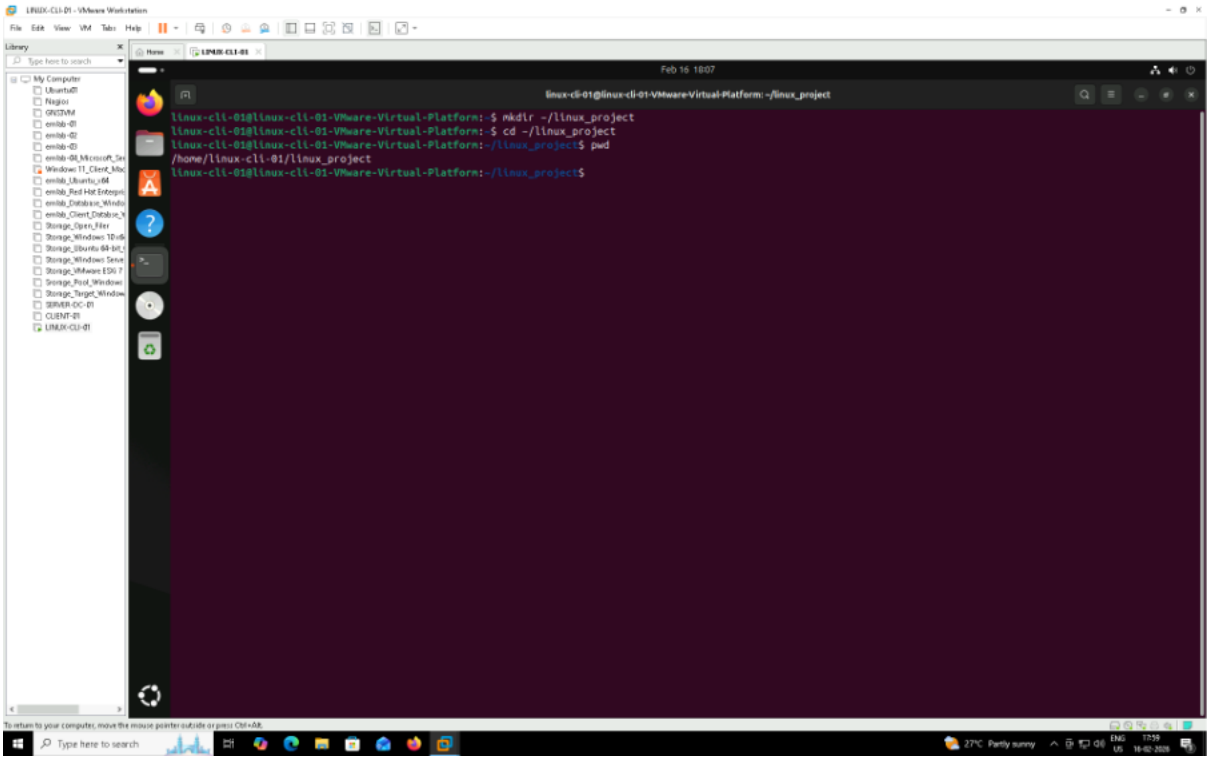




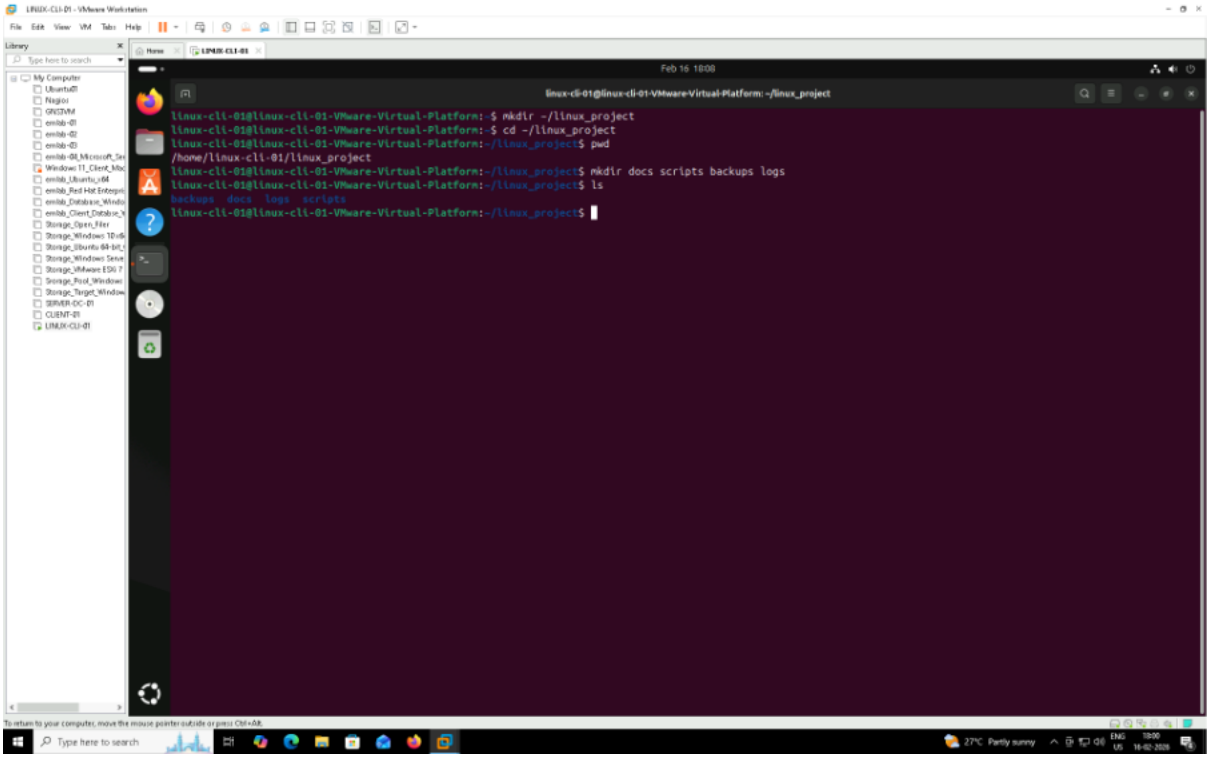


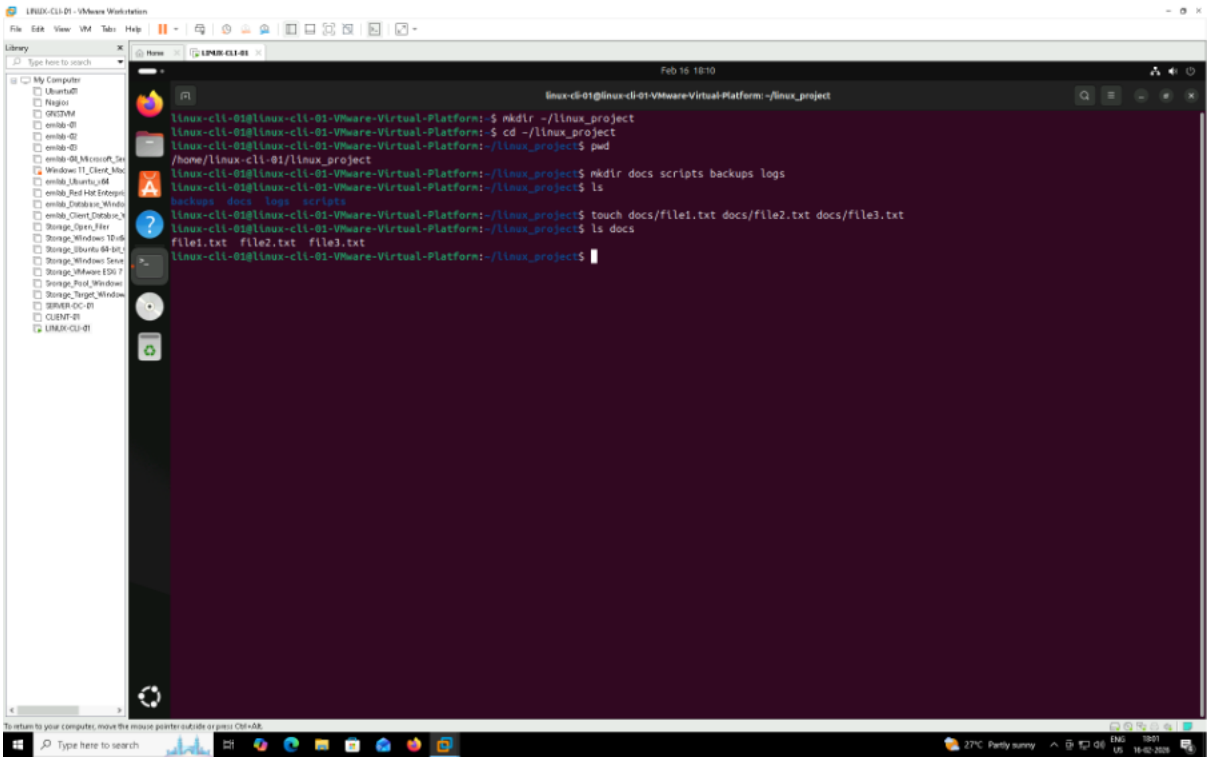
The Linux Environment was verified using basic system identification commands to confirm user and working directory.

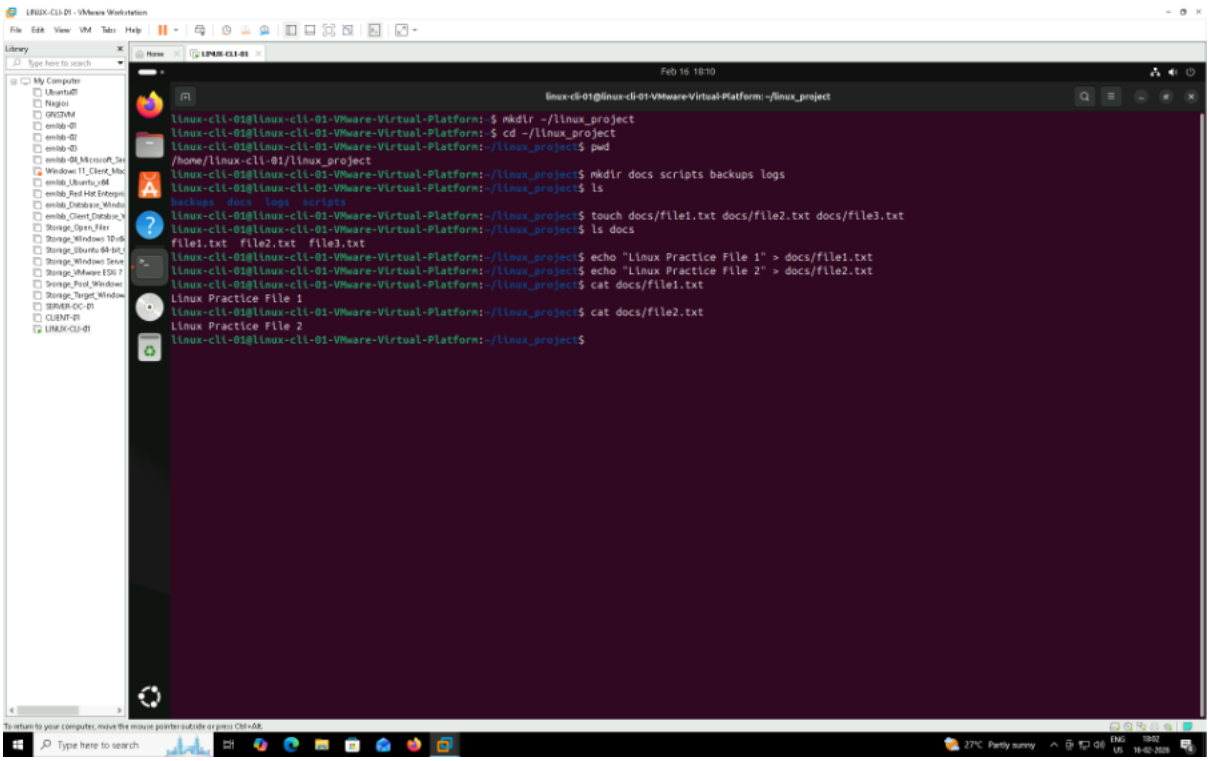
A personal working directory named linux_project was created inside the user home directory to store all the practice files.



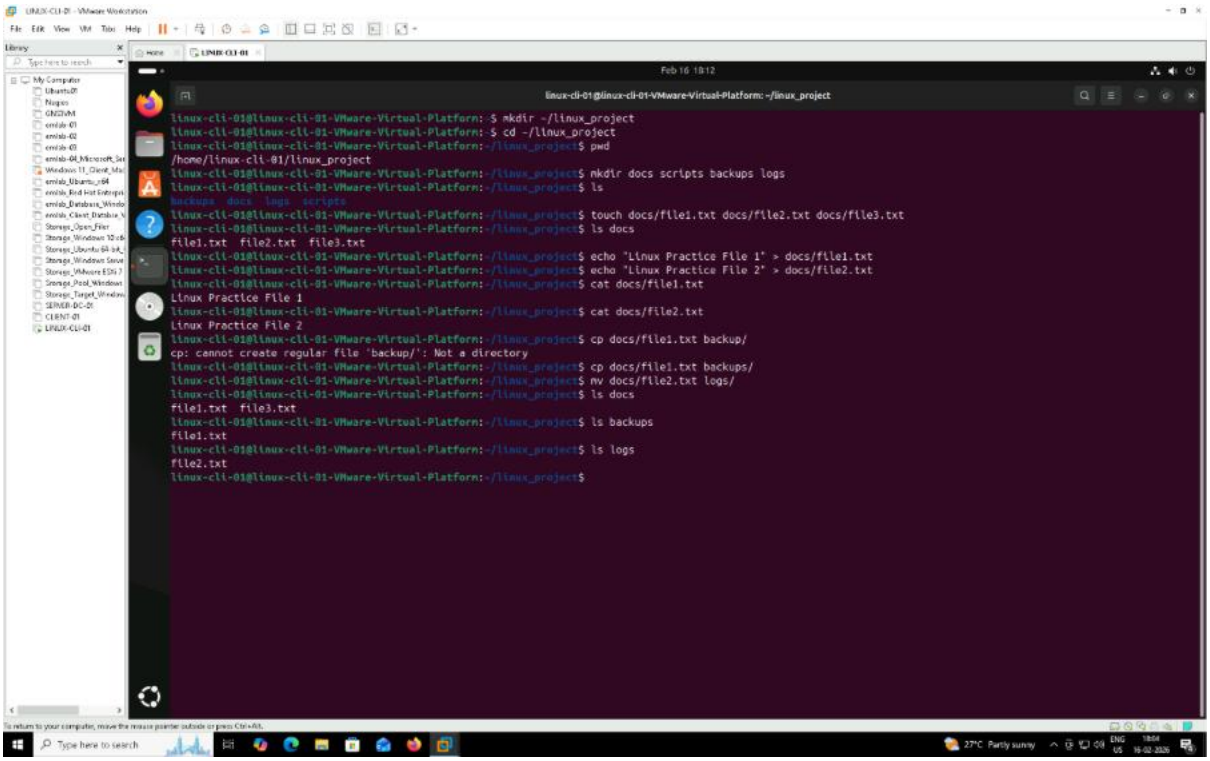
Multiple subdirectories were created to organize files based on usage such as documentation, scripts, backups and logs.


Feb 16 18:09
linux-cli-01@linux-cli-01-VMware-Virtual-Platform: ~/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~$ mkdir ~/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~$ cd ~/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ pwd
/home/linux-cli-01/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ mkdir docs scripts backups logs
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ ls
backups docs logs scripts
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$

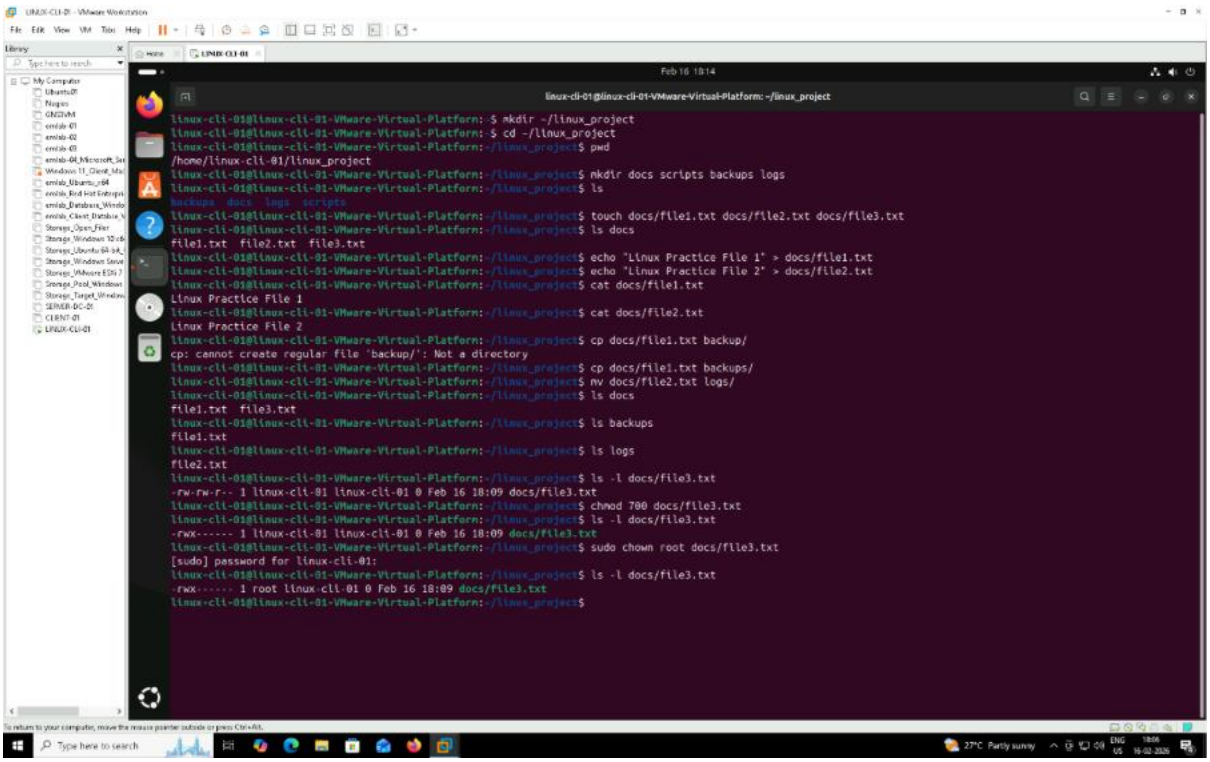

Feb 16 18:10
linux-cli-01@linux-cli-01-VMware-Virtual-Platform: ~/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~$ mkdir ~/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~$ cd ~/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ pwd
/home/linux-cli-01/linux_project
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ mkdir docs scripts backups logs
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ ls
backups docs logs scripts
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ touch docs/file1.txt docs/file2.txt docs/file3.txt
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$ ls docs
file1.txt file2.txt file3.txt
linux-cli-01@linux-cli-01-VMware-Virtual-Platform:~/linux_project$



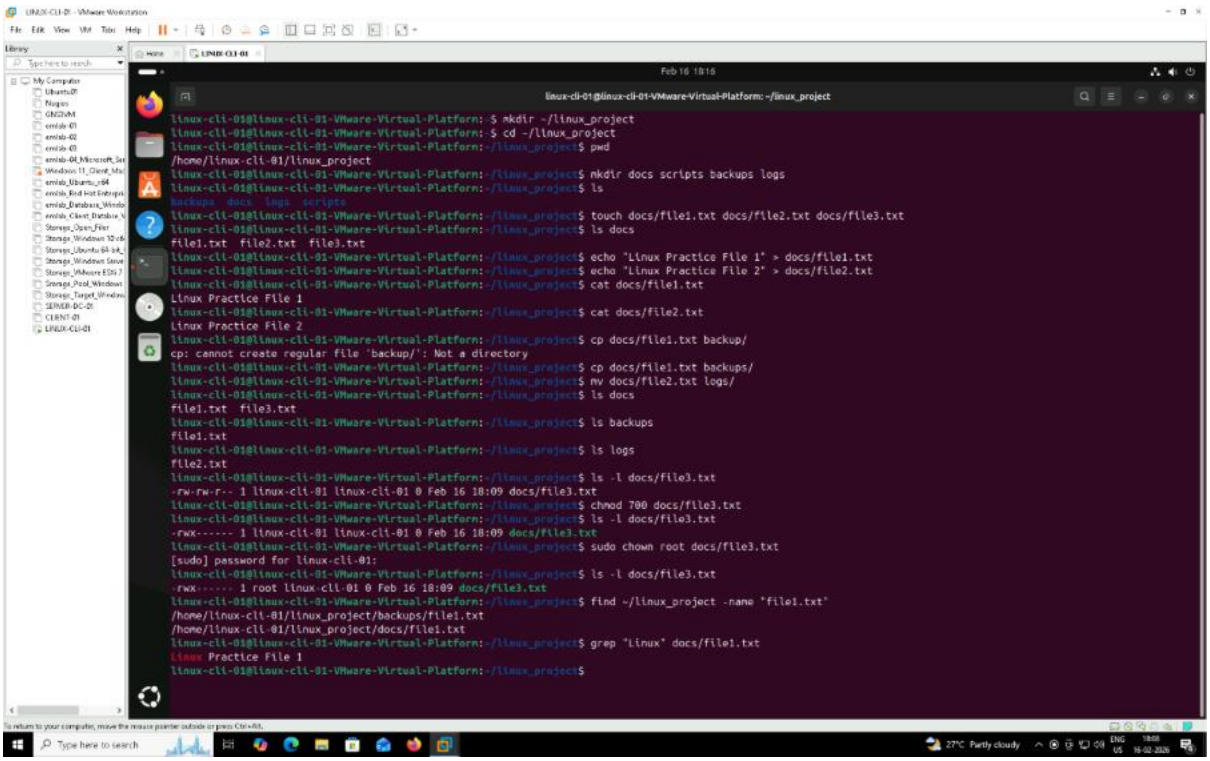
Text files were created using the touch command and populated with sample data using echo redirection.



Files were copied and moved between directories using cp and mv commands to demonstrate file management.



File permissions were modified using chmod and ownership was changed using chown to demonstrate access control in Linux.

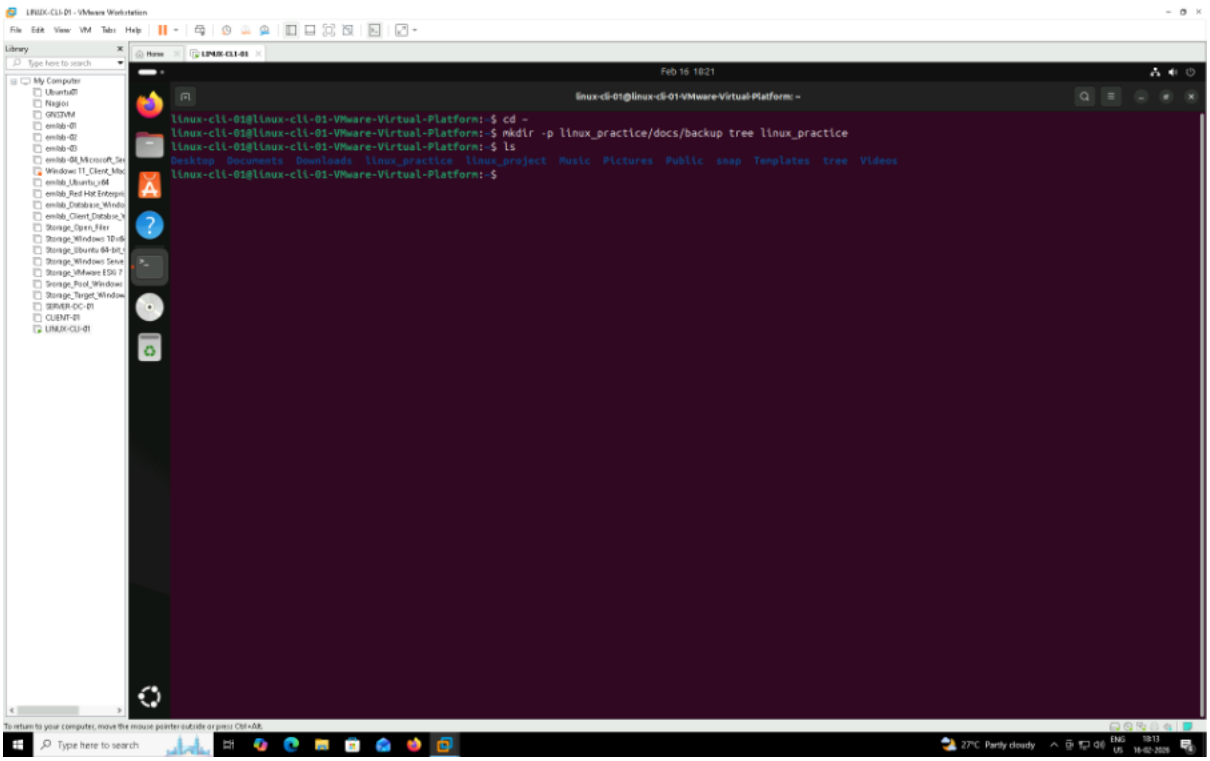


The find command was used to locate files and grep was used to filter specific content within files.

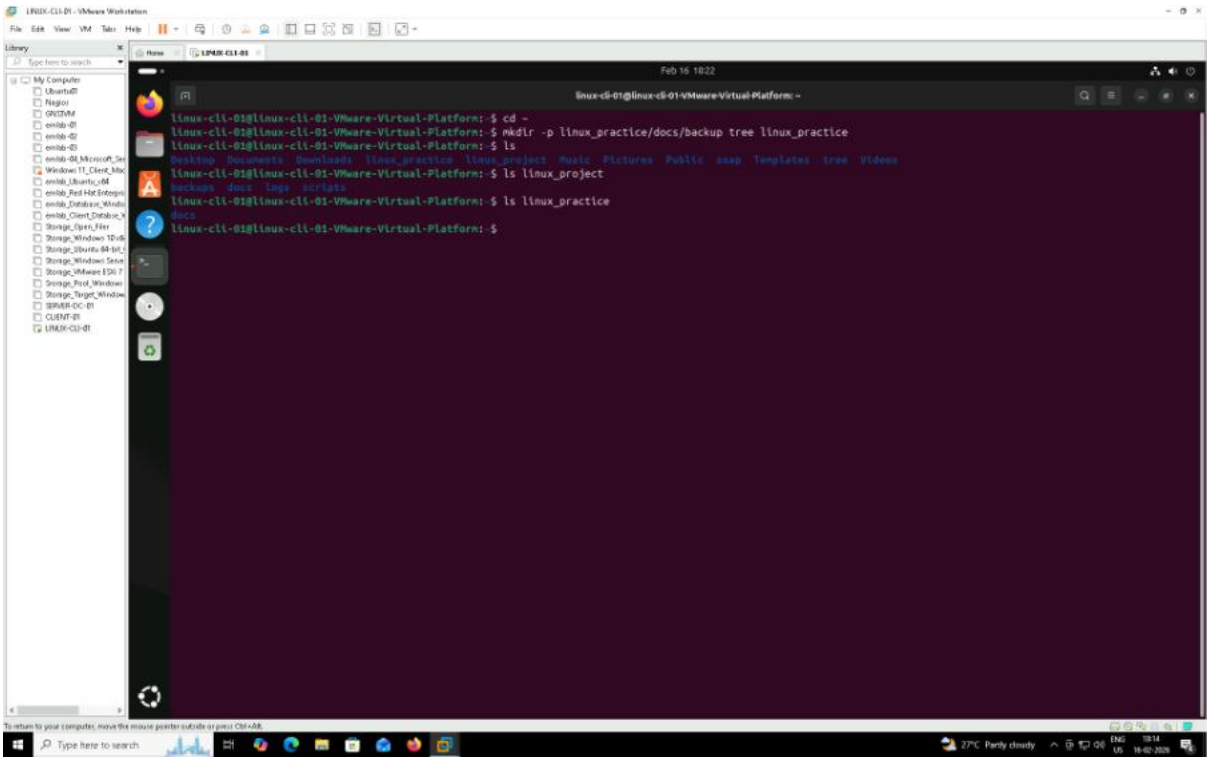
Assignment 2: Directory and File Management

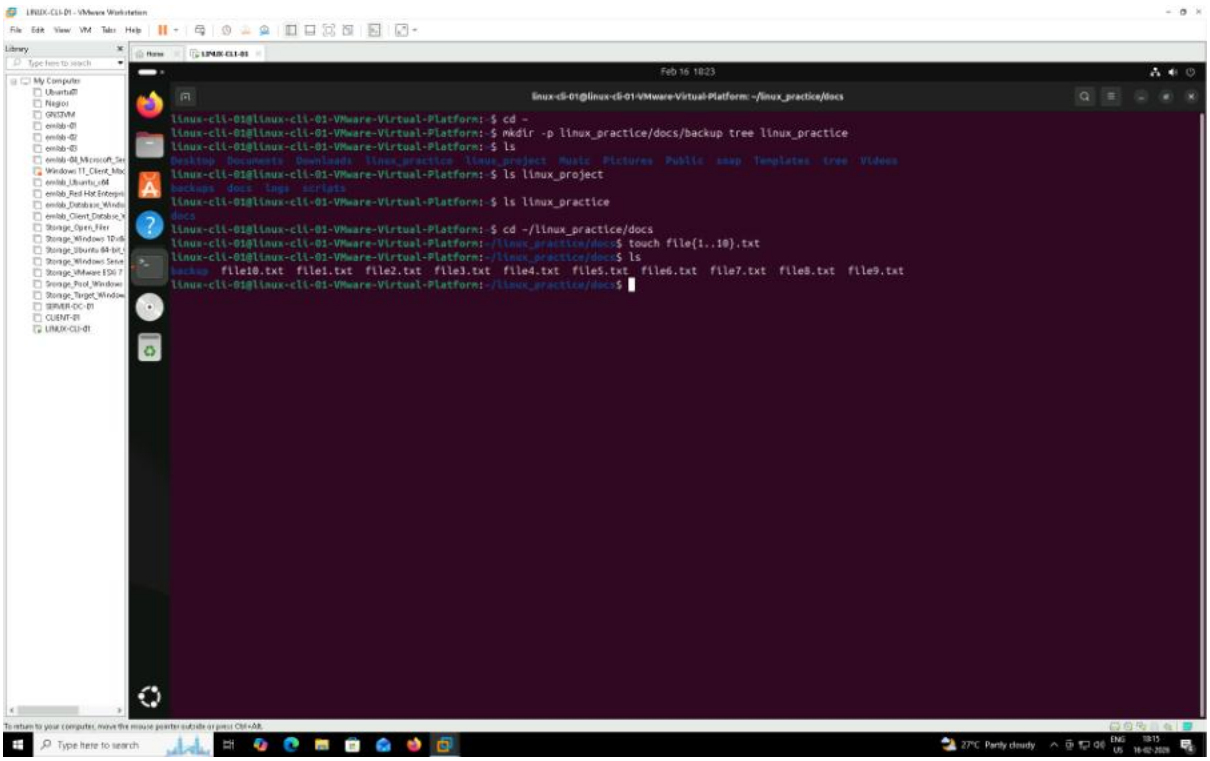
As part of this assignment, you need to complete below tasks:

1. Create a directory structure: /home/student/linux_practice/docs/backup
2. Create 10 empty text files with sequential names
3. Copy files between directories
4. Move files using relative and absolute paths
5. Delete specific files and empty directories

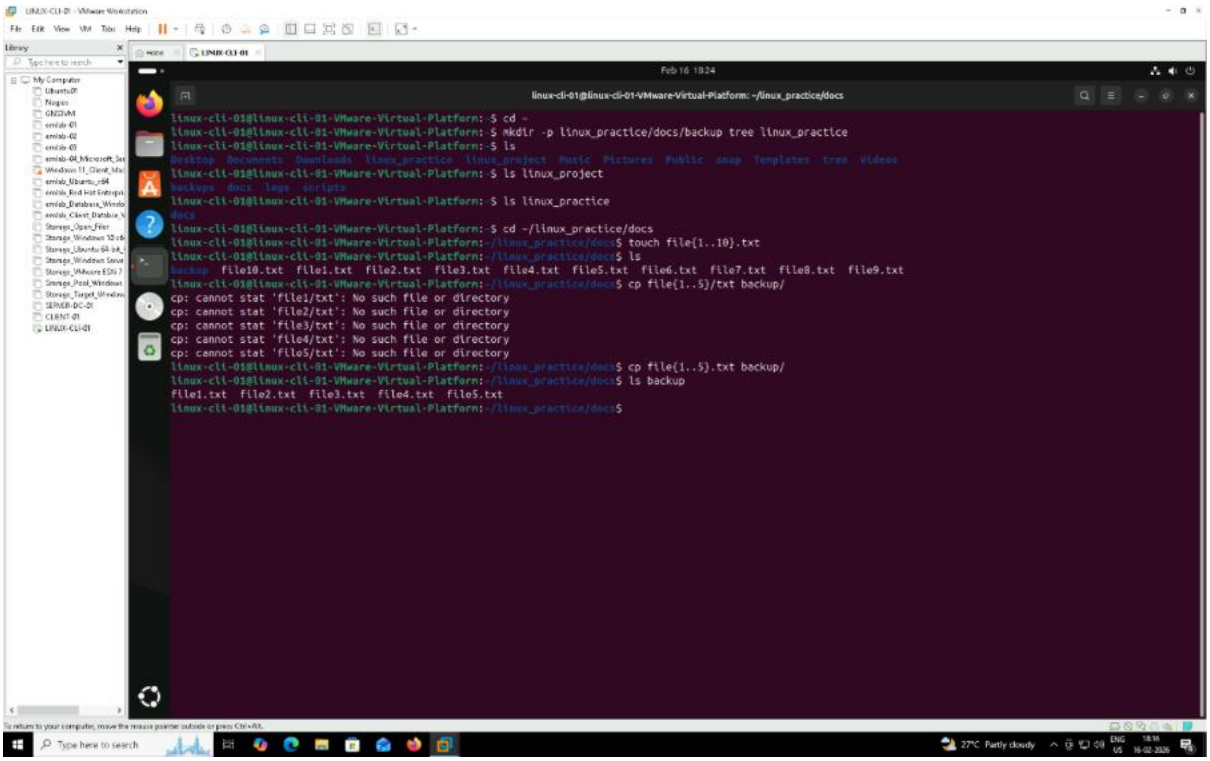


A nested directory structure was created using the mkdir -p command to automatically generate parent and child directories in a single operation.

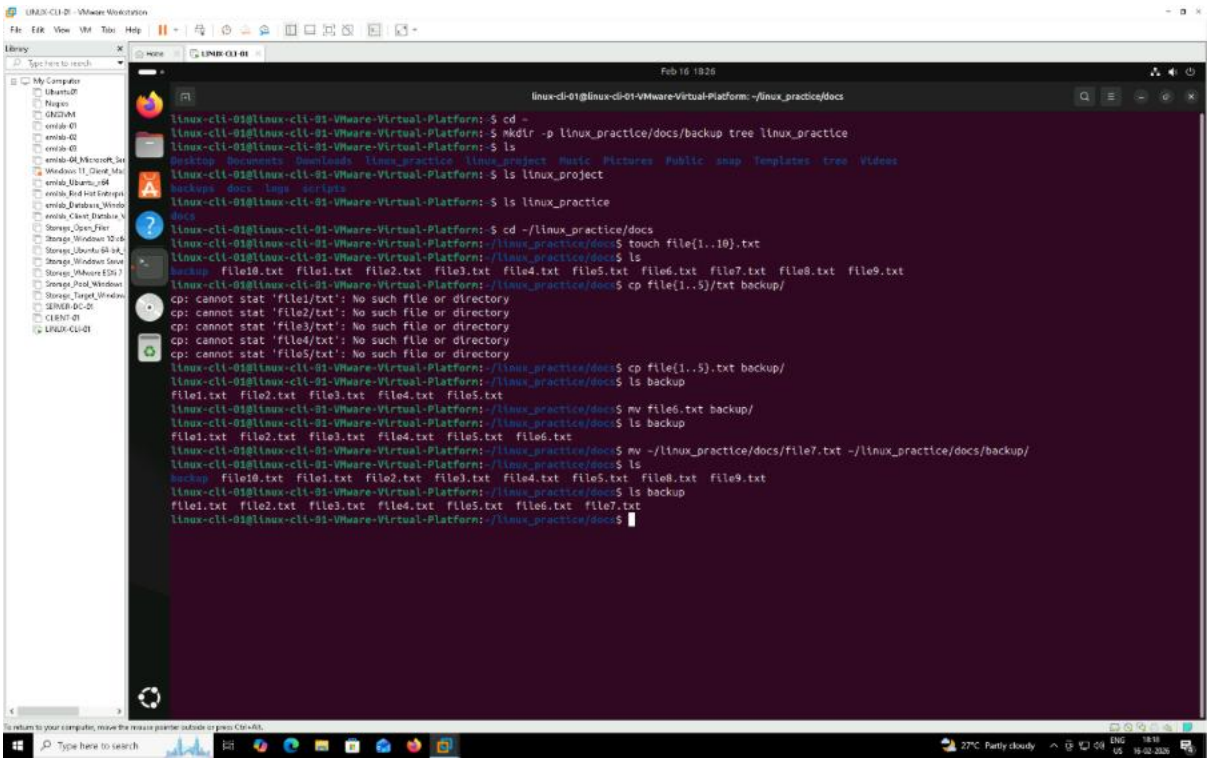




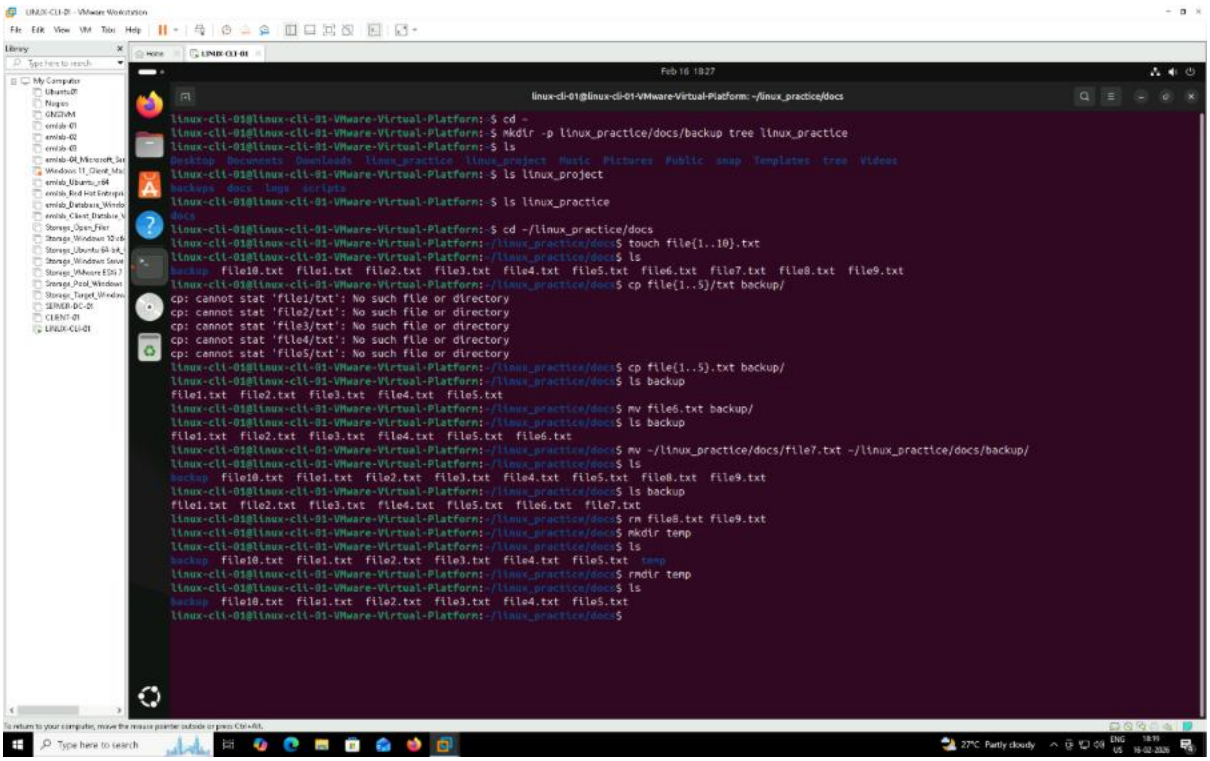
Ten empty text files were created using brace expansion with the touch command to demonstrate batch file creation.



Selected files were copied to a backup directory using the co command to simulate data backup operations.



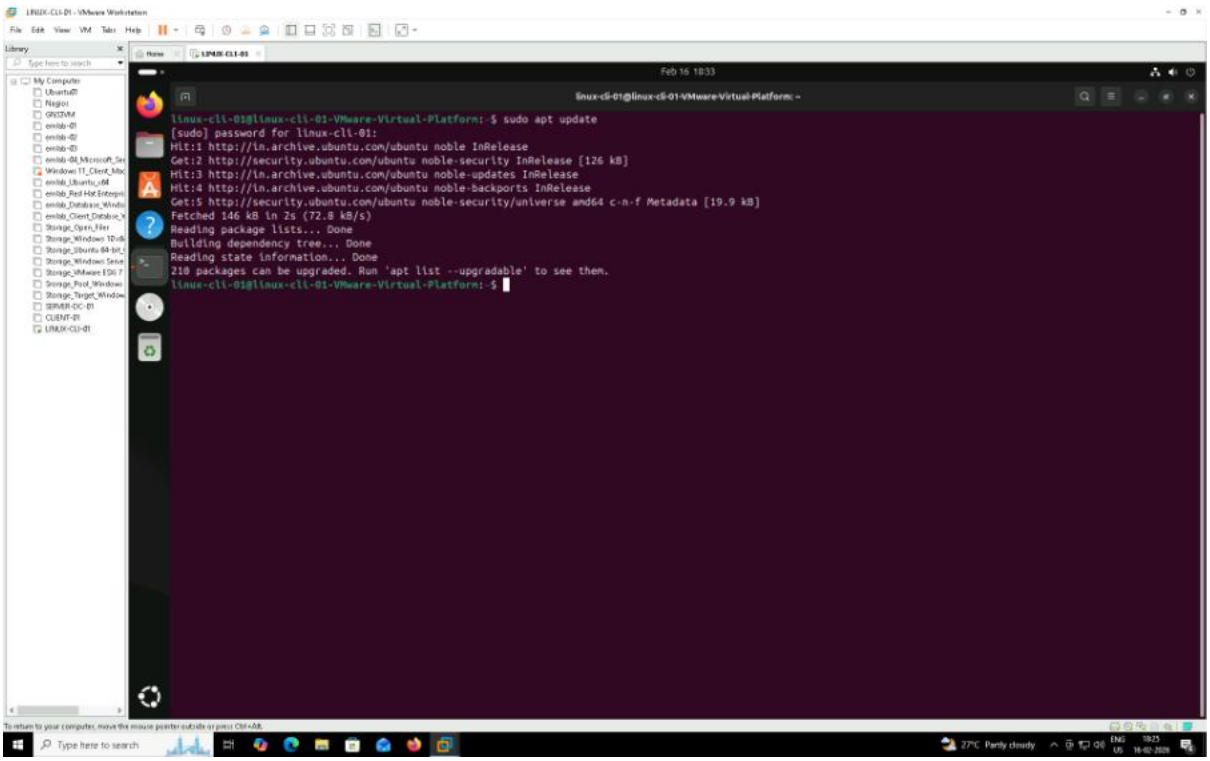
Files were moved using both relative and absolute paths to demonstrate different file referencing methods in Linux.

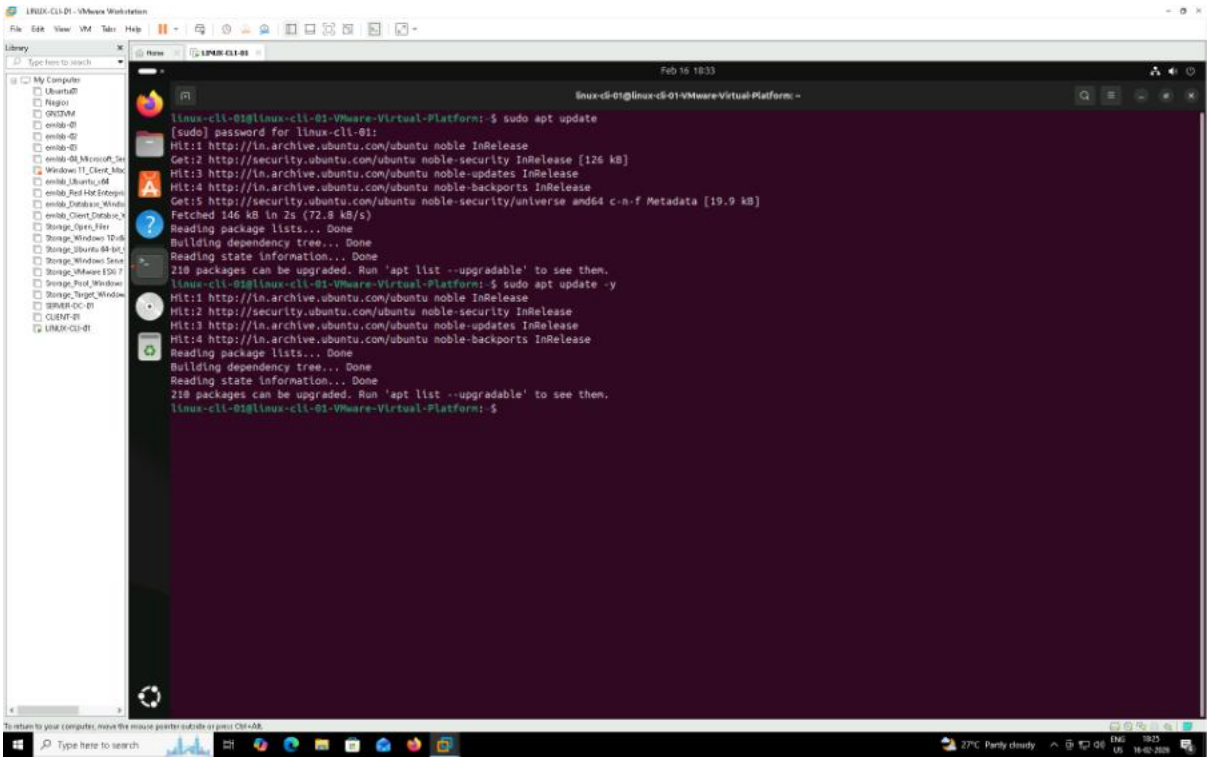


Files were removed using rm and empty directories were deleted using rmdir to demonstrate safe file system cleanup.

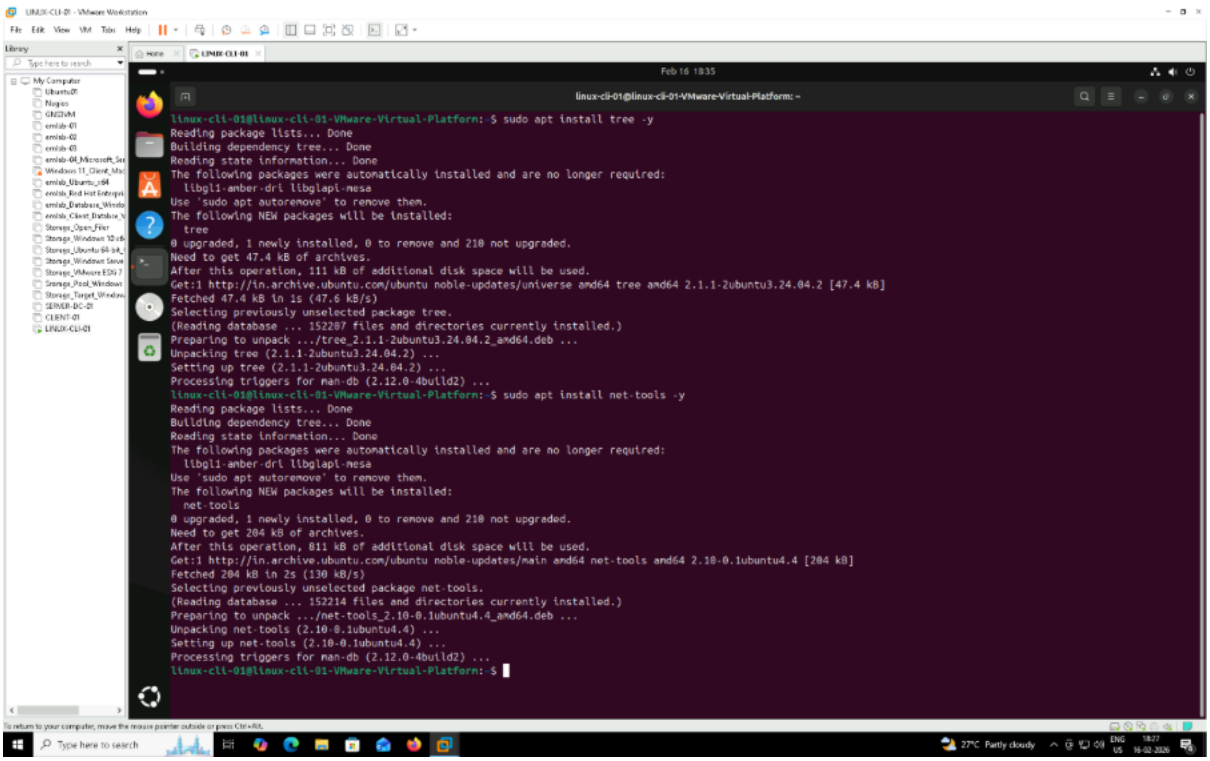
Assignment 3: Package Management

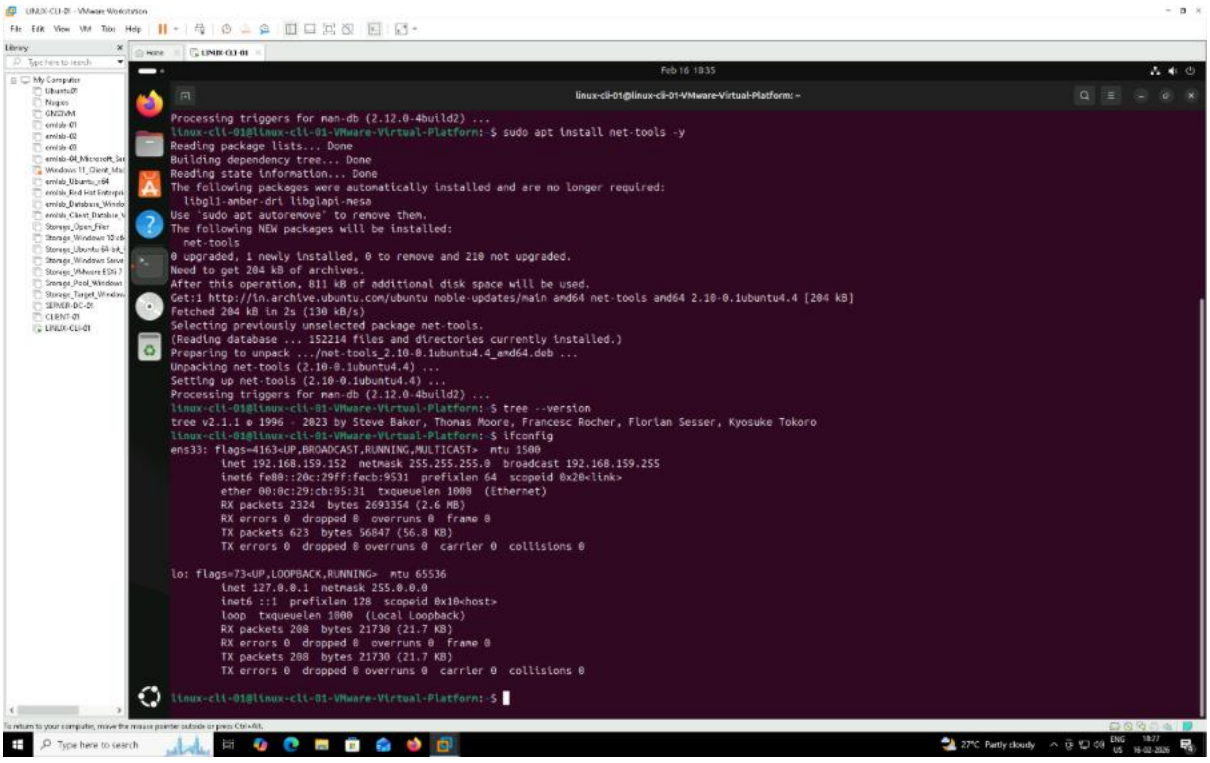
1. As part of this assignment, you need to complete below tasks:
2. Update system packages
3. Install software from repositories
4. Remove unwanted packages
5. Create local package repositories
6. Manage package dependencies



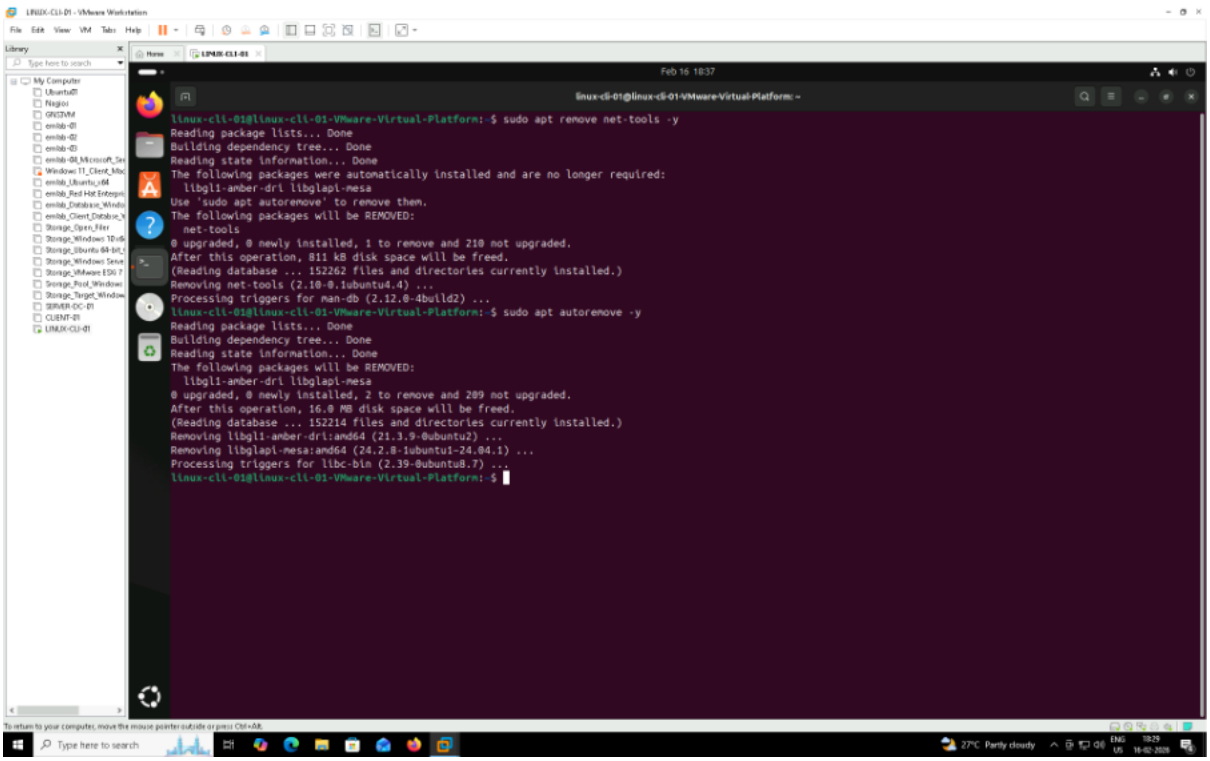


The package index was refreshed and installed packages were upgraded using the apt package manager to ensure the system contained the latest security and software updates.

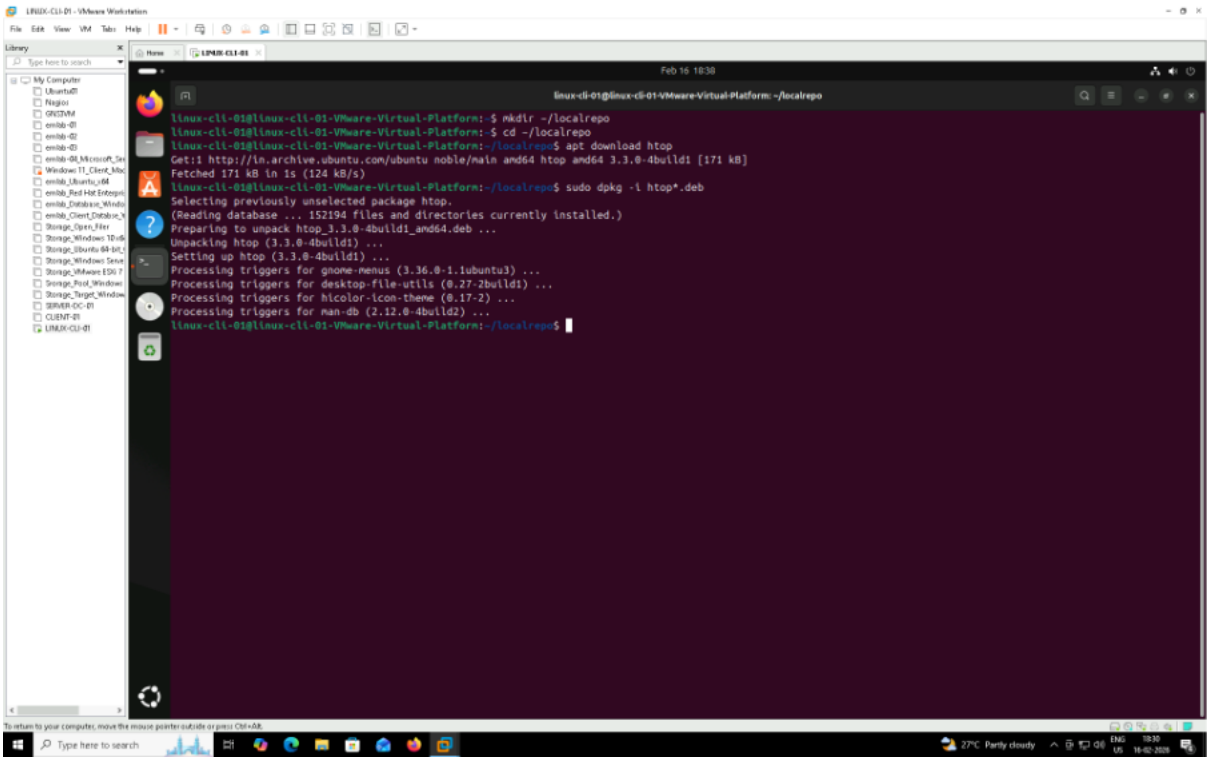




Software packages were installed from the default repository using the apt install command and verified after installation.

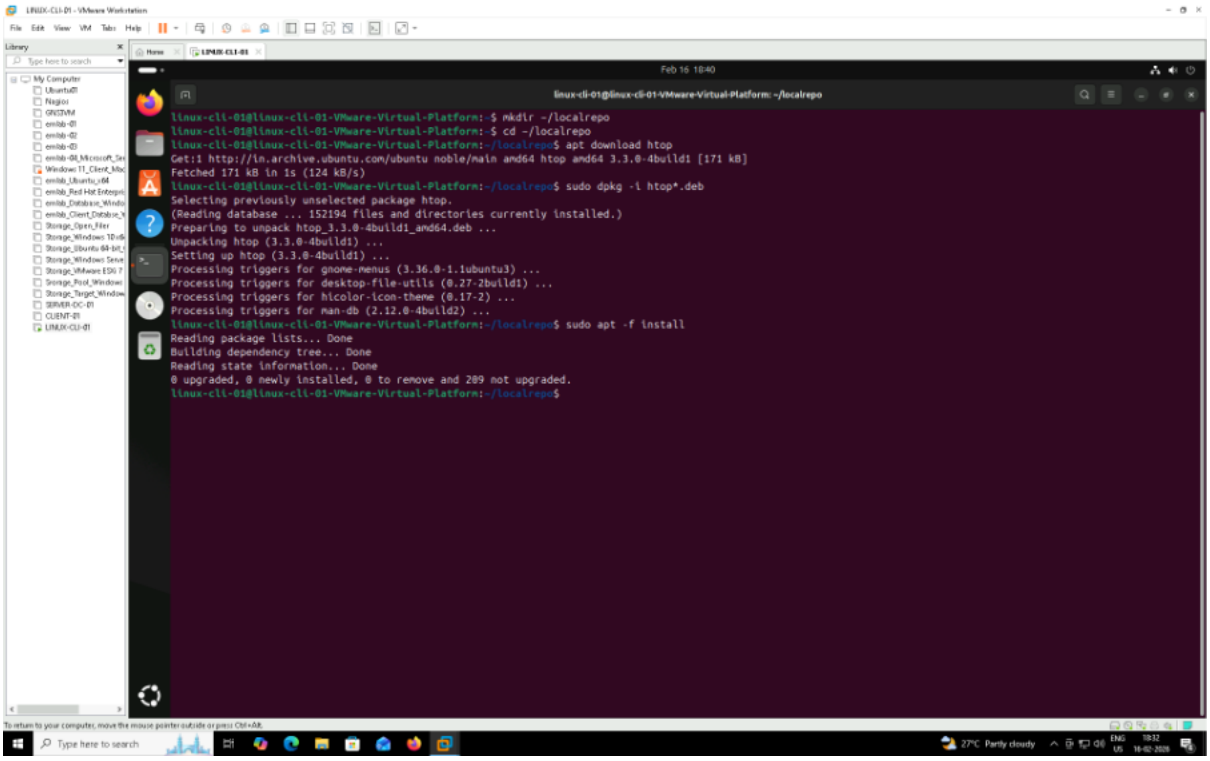


Unnecessary packages were removed using apt remove and unused dependencies were cleared using autoremove.



A local package file was downloaded and installed manually using dkpg to simulate installation from a local repository source.

Sometimes dkpg fails due to missing dependencies.



Missing package dependencies were automatically resolved using the apt -f install command.

Conclusion: -

The Linux assignment provided hands-on experience in system administration and command-line operations within a Linux environment. Core concepts such as file system navigation, user and group management, file permissions, package management, and basic networking configuration were implemented. Essential commands and administrative tools were used to manage system resources and troubleshoot issues effectively.

The exercises enhanced understanding of Linux architecture, process management, and basic networking configuration were implemented. Essential commands and administrative tools were used to manage system resources and troubleshoot issues effectively.

Overall, the Linux practical sessions strengthened system administration skills and built confidence in managing Linux0based systems in real-world IT environments.